THEORY PART

Total Marks: **8** Duration: 20 **mins**

I. **TRUE/FALSE**. (0.5 * 2 = 1 Mark)

Underline **T** if the statement is correct, otherwise Underline **F**.

1. HTTP protocol governs the way a web Server and web Client communicates.	T / F
2. Ping command is used to verify telnet connectivity from a PC to a Switch.	T / F

II. **MULTIPLE CHOICES**. (0.5 * 3 = 1.5 Marks)

Encircle the letter of the correct answer that corresponds to the given statement.

3. What is the length of an IPV4 address?

   a) 128 bits
   b) 32 bits
   c) 64 bits
   d) 48 bits

4. What is the measure of usable data transferred over a given period of time?

   a) Bandwidth
   b) Throughput
   c) Goodput
   d) Latency

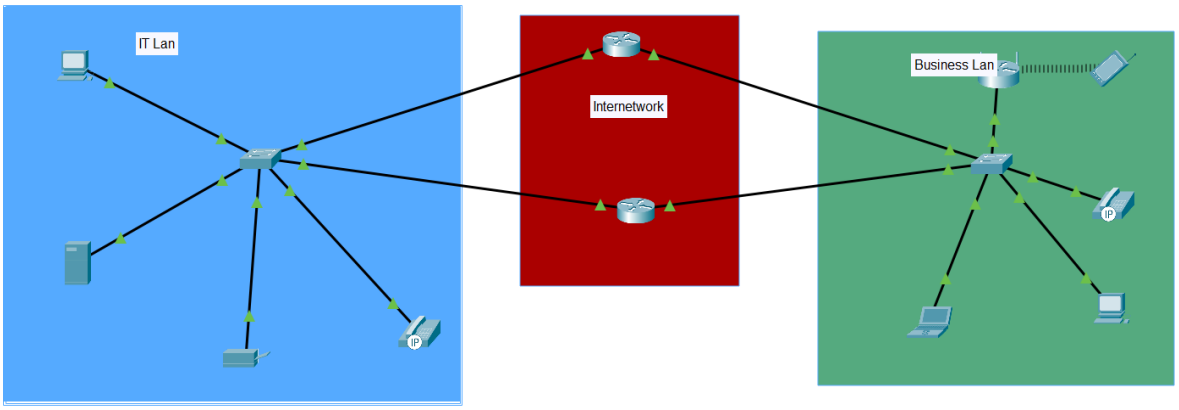
5. What is the function of Unicast message delivery option?

   a) It delivers message from one end device to a group of end devices.
   b) It delivers message from one end device to all end devices.
   c) It delivers message from a group of end devices to another group.
   d) It delivers message from one end device to another end device.

III. **SHORT ESSAY**. (5.5 Marks)

Answer all the questions briefly and clearly.

6. Refer to the exhibit and identify the number of end devices. (1.0 Mark)



How many **end devices** are there in the topology? _____

List down the **names of end devices** (do not write the name of same end device)

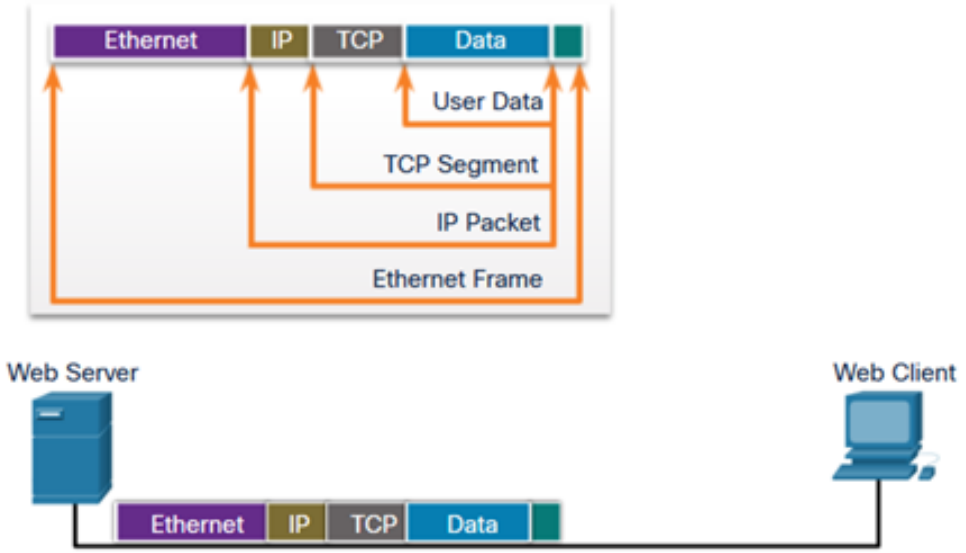
7. Why would you recommend **Copper cable** for your company network?(any 2 reasons)

(1.0 Mark)

8. What is the function of **Segmentation** in networking? How does it increase the speed of network?

(1.5 Marks)

9. The diagram shows the encapsulation process between a Server and Client. (2 marks)



a. Analyse the diagram and briefly explain the **Encapsulation** process.

b. List down the **four (4) layers of the TCP/IP** Reference **Model**.

PRACTICAL PART **Duration: 40 minutes**

Total Marks: **12**

Name: _____ ID Number: _____

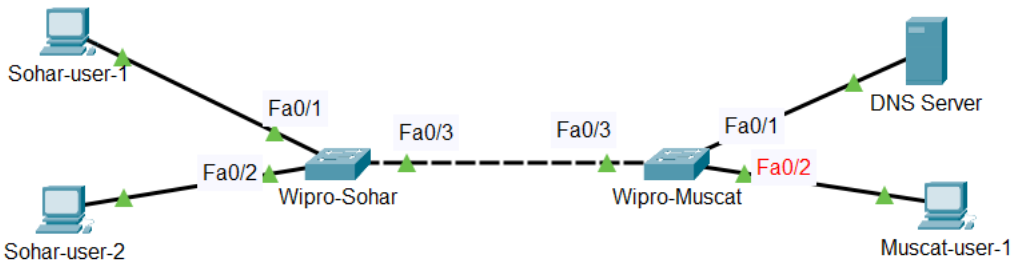
Section: _____ Computer Number: _____

General Instructions:

- Create a folder in **DESKTOP** and name it as **ME_StudentID_Name**. Example: M**E_S01234_Ali**
- **SAVE** your Packet tracer file in the created folder.
- Please **DON'T LOG-OFF** or **SHUTDOWN** your computer after the examination to avoid losing your file/s.
- Once done copy the folder to **Drive Z**

Answer the following questions.

10. **Create** the topology given below in the Cisco Packet Tracer. (2.0 Marks)



Addressing Table:

Device Name	Interface	IP Address	Subnet Mask
Sohar-user-1	Fa0	192.168.50.1	255.255.255.0
Sohar-user-2	Fa0	192.168.50.2	255.255.255.0
DNS Server	Fa0	192.168.50.3	255.255.255.0
Muscat-USER-1	Fa0	192.168.50.4	255.255.255.0
Wipro-Sohar Switch	VLAN 1	192.168.50.5	255.255.255.0
Wipro-Muscat Switch	VLAN 1	192.168.50.6	255.255.255.0

11. **Configure** the IP addresses of Sohar-user1, Sohar-user-2, DNS-Server and Muscat-user-1 PCs by taking the address from the above addressing table. (2.0 Marks)

12. **Configure** the **Wipro-Sohar Switch** with the following configurations. (4.0 Marks)

- Configure the **hostname** as Wipro-Sohar.
- Assign the **banner** as "This Switch is configured by Student Name".
- Configure the **unencrypted Privilege level password** as **Student ID**.
- Configure the **Telnet password** as **Student first name**.
- Configure **VLAN1 interface** of Wipro-Sohar Switch with IP address from the above addressing table and enable it.

  Save the running configuration file to start up configuration file.

13. **Configure** the **Wipro-Muscat** Switch with the following configurations. (3 Marks)

- Configure the **Console password** as wipro123.
- **Encrypt** all plain text passwords.
- Configure **VLAN1 interface** of Wipro-Muscat switch with IP address from the above addressing table and enable it.

  Save the running configuration file to start up configuration file.

14. **Analyze** the given output and Write down the **command which display the output** (1 Mark)

```
Vlan1 is up, line protocol is down
  Hardware is CPU Interface, address is 0060.3e77.1371 (bia 0060.3e77.1371)
  Internet address is 194.168.70.5/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 21:40:21, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     1682 packets input, 530955 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicast)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     563859 packets output, 0 bytes, 0 underruns
     0 output errors, 23 interface resets
     0 output buffer failures, 0 output buffers swapped out
```

Answer:_____